The Compute Merit Order: Where AI Compute Is Economic to Run

Compute Merit Order Project

Working paper — draft, August 3, 2026

Abstract

A GPU cluster earns a compute price per GPU-hour and pays an electricity cost per GPU-hour; the difference is a spread directly analogous to the spark spread long used in power markets. We propose that this compute spark spread determines a shutdown price below which a given fleet is uneconomic to run, and that because industrial electricity costs vary by roughly a factor of three across regions, shutdown prices should vary similarly — implying a predictable global merit order in which capacity retires by geography as compute prices fall. We further ask whether the binding constraint on AI data-centre buildout is grid interconnection and power availability rather than chip supply. This is a working paper: the present draft documents methodology and reports results only for the data sources brought online so far. *[Pending: aggregate results once more hubs are populated; see Section 8]*.

Contents

1 Introduction

Spark spreads — the difference between the wholesale price of electricity and the fuel cost of generating it — have been a standard tool in power-market economics for decades, used to determine when a gas-fired plant is worth dispatching. We argue an analogous quantity governs AI compute: a GPU cluster sells compute capacity at some market price per GPU-hour, and its main variable cost is the electricity it consumes. When the compute price falls below the cost of power (plus non-power marginal operating costs), running the fleet no longer covers its variable cost, and rational operators curtail or shut it down.

This became analysable in general, rather than case-by-case, only once two things existed together: (a) reasonably transparent, high-frequency regional wholesale electricity price data, which has existed for years in most liberalised power markets, and (b) a visible compute price — i.e., published GPU rental rates and compute-price indices that let one observe the revenue side of the spread at all. The latter only began to be published with any regularity in 2026, which is why this analysis is being done now rather than earlier.

We test two claims in this paper:

1. **The compute merit order.** Because industrial power costs vary by roughly 3x across the regions we study, shutdown prices (Section 3) should vary similarly, producing a predictable geographic sequence in which capacity retires as compute prices fall (Section 5).

2. **Chips vs. power.** The binding constraint on AI data-centre buildout is grid interconnection and power availability, not chip supply (Section 6).

Both claims are falsifiable from the data assembled here, and we report where the data does not support them rather than adjust the claims to fit. As of this draft, only the ENTSO-E DE-LU (Germany/Luxembourg) bidding zone has real data behind it; every other hub is a documented placeholder. *[Pending: full cross-hub test of both claims once Phase 1–4 data collection completes; see the project's GitHub issue tracker for status]*.

2 Data and methods

2.1 Sources

Every value in this paper and its accompanying dataset carries a `source_url` and `as_of_date`; the full source list and retrieval methodology live in `docs/SOURCES.md` and `docs/METHODOLOGY.md` in the project repository, and are summarised here. Power price data for European bidding zones comes from the ENTSO-E Transparency Platform (ENTSO-E, 2026); US ISO data (not yet pulled) is planned via the `gridstatus` library (gridstatus contributors, 2026); currency conversion uses ECB Statistical Data Warehouse monthly reference rates (European Central Bank, 2026); interconnection-queue data is planned from LBNL's Queued Up dataset (Lawrence Berkeley National Laboratory, 2026).

2.2 Confidence levels

Every hub in `data/final/hubs.geojson` (Section **??**, not yet populated in this draft) will carry a `confidence` field of `high`, `medium`, or `low`. A hub with no open API (e.g. Gulf states, China, Malaysia, India as of this draft) is represented as a null row with `confidence=low` and a linked GitHub Issue describing what would resolve it, rather than an estimated figure.

Hub	Confidence	Basis
DE-LU (Germany/Luxembourg)	Pipeline built, data pending	ENTSO-E Transparency Platform; requires a free API token not yet provisioned in this environment (see `docs/SOURCES.md`)
All other hubs	Not started	Backlogged as GitHub Issues; see Section 8

Table 1: Confidence status by hub, this draft.

3 The compute spark spread

For a GPU fleet of nameplate accelerator power draw P_{TDP}, define the delivered power per accelerator-hour as

$$\text{delivered\_kW} = \mathrm{PUE} \times (P_{\mathrm{TDP}} + P_{\mathrm{overhead}})\,, \tag{1}$$

where P_{overhead} is the CPU/NIC/storage share attributable to one accelerator. This project fixes a single named default of 1.10 kW per GPU-hour (a 700 W accelerator TDP at PUE 1.25, plus shared overhead) as a config constant, and reports sensitivity across 0.9 kW–1.4 kW (Section 7).

The power cost per GPU-hour is then

$$\text{power\_cost} = \text{delivered\_kW} \times \text{price}_{\mathrm{USD/kWh}}. \tag{2}$$

The *shutdown price* adds non-power marginal operating expense (staffing, bandwidth, maintenance), which we deliberately parameterise as a range rather than a point estimate:

$$\text{shutdown\_price} = \text{power\_cost} + \text{opex}_{\mathrm{marginal}}. \tag{3}$$

The compute spark spread and headroom are

$$\text{headroom} = \frac{\text{compute\_price} - \text{shutdown\_price}}{\text{compute\_price}}. \tag{4}$$

[Pending: worked numeric example once DE-LU data and a compute-price series are both in `data/final`].

4 Regional cost surfaces

[Pending: per-hub power price tables and figures once more than one hub has real data — this draft has only DE-LU pipeline-ready, with no pulled values yet].

5 The merit order

The central deliverable of this project is a merit-order curve: all tracked capacity sorted ascending by marginal cost per GPU-hour, against which a reader can compare a compute-price line to see what capacity is uneconomic at that price. *[Pending: this figure requires shutdown prices for multiple hubs and is not yet computable from a single pipeline; see `models/` backlog].*

6 Constraint analysis: chips vs. power

[Pending: interconnection-queue and grid-headroom data collection is Phase 2 of this project and has not started as of this draft; see open GitHub issues].

7 Sensitivity

We commit in advance to varying delivered_kW, non-power opex, PUE, and compute price, and reporting where the two central claims (Section 1) break down rather than merely where they hold. *[Pending: sensitivity tables once Section 3's inputs are populated with real data].*

8 Limitations

This draft is intentionally incomplete, and we list every gap rather than smooth over it:

- **No pulled data yet.** The ENTSO-E DE-LU pipeline (`pipelines/entsoe.py`) is implemented and tested against a synthetic fixture, but has not been run against the live API: it requires a free `ENTSOE_API_TOKEN` that was not available in the environment this draft was produced in. See `docs/SOURCES.md` for registration steps.
- **Ten of twelve hubs are not started.** Only DE-LU has a pipeline; the remaining European zones, US ISOs, Singapore, and the document-derived hubs (Gulf states, China, India) are tracked as GitHub issues, not estimated.
- **GPU counts per site are not disclosed** by operators in general, so installed/announced capacity figures will rely on third-party estimates (CBRE, DC Byte) with their own uncertainty.
- **PUE is self-reported** by operators where published at all, and the 1.25 figure used as this project's default is an industry-typical value, not a measurement.
- **PPA (power purchase agreement) prices are confidential** in most jurisdictions; this project uses public wholesale/day-ahead prices as a proxy, which will differ from actual contracted rates, typically understating the compute operator's true cost certainty (and sometimes its price level) in either direction depending on contract structure.
- **Gulf and China data, when added, will be document-derived** (government/TSO reports, not live APIs), and will carry `confidence=low` accordingly.
- **The compute-price leg is the weakest input, and the asymmetry is structural.** A survey conducted 2026-08-03 (`docs/SOURCES.md`) found no free, redistributable compute-price index of comparable quality to the power data. The two credible indices identified — Silicon Data's `SDH100RT` and GetDeploying's GPU rental price index — are both paywalled, and the latter's terms explicitly prohibit redistributing raw data feeds, which is incompatible with publishing a CC-BY-4.0 dataset. The cost leg of this spread therefore rests on open regulatory data while the revenue leg rests on proprietary commercial data. We intend to construct a transparent first-party index from published provider price pages instead; until then, every headroom figure inherits the weaker leg.

- **On-demand list price is not the realised price.** Large operators transact on multi-year reserved contracts at material and confidential discounts. This is the exact mirror of the PPA problem on the power side: *both* legs of the spread are public proxies for confidential real transactions.
- **Ordering and level have different confidence.** Which hub goes uneconomic first depends mainly on the power leg and is comparatively robust; the absolute compute price at which it happens depends on the weak leg. We hedge these two claims differently throughout, and readers should not transfer confidence from the former to the latter.

9 Conclusion

[Pending: to be written once Section 5 and Section 6 have real results to conclude from. This project is under active, multi-session construction; see the GitHub repository's Issues and Milestones for current status rather than a prose summary here, per the project's own contribution rules.]

References

ENTSO-E. ENTSO-E transparency platform. `https://transparency.entsoe.eu/`, 2026. Day-ahead electricity price data by bidding zone, accessed 2026.

European Central Bank. ECB statistical data warehouse: euro foreign exchange reference rates. `https://sdw.ecb.europa.eu/`, 2026. Monthly average EUR reference rates, accessed 2026.

gridstatus contributors. gridstatus: a python library for iso/rto electricity market data. `https://github.com/gridstatus/gridstatus`, 2026. Accessed 2026.

Lawrence Berkeley National Laboratory. Queued up: Interconnection queue data. `https://emp.lbl.gov/queues`, 2026. Generation and storage interconnection queue dataset, accessed 2026.